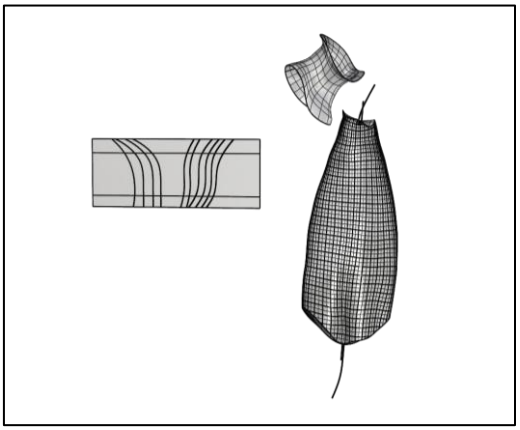
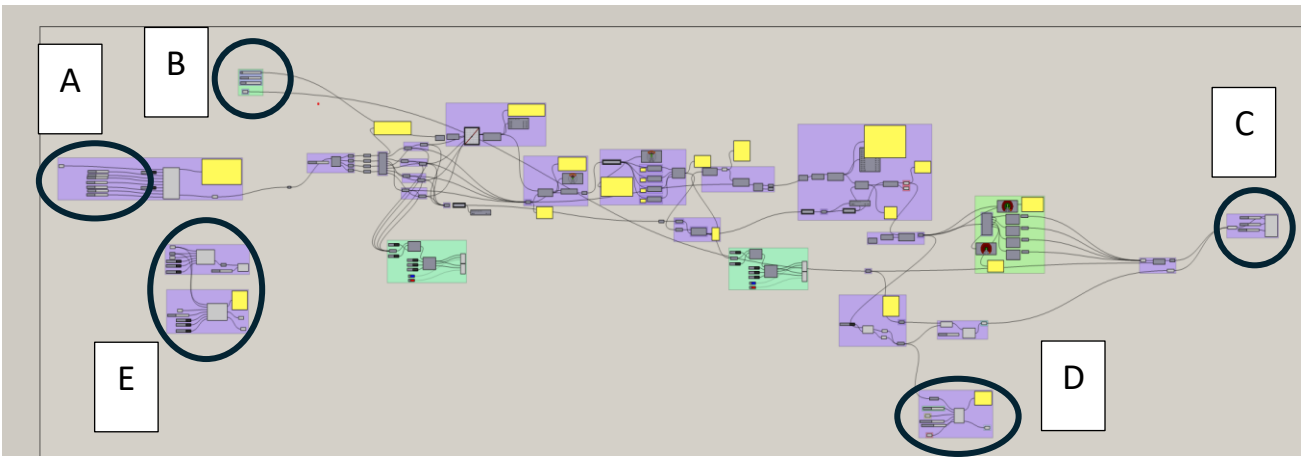
This document explains how to use the parametric design program files submitted as auxiliary supplementary materials. These materials include programs for generating muscle patterns and ligament patterns, as well as examples of their usage. The supplementary materials are organized as follows:

- example.3dm
  - A collection of example models for muscles and ligaments.
- pattern_generator.gh
  - A collection of programs for generating muscles and ligaments. The file example.3dm contains geometries divided into layers, as shown in the following figure, for use in the sample program.



The pattern_generator.gh is composed of the following components.
Note that the sections without numbers represent the pattern generation program.
The numbered components are referenced as **A** through **E**.
All input parameters are adjusted to values suitable for the sample model.



1. Muscle Generation

To generate muscles, example input volumes are provided in the example.3dm file.

Parent layer: **muscle**

- **lattice curve** layer: Curves that serve as the basis for lattice coordinates.
- **muscle_shape_with_wire** layer: The external shape of the muscle with the wire outline removed.
- **wire_curve** layer: Curves indicating the positions of the wires.
- **muscle_shape** layer: The external shape of the muscle.

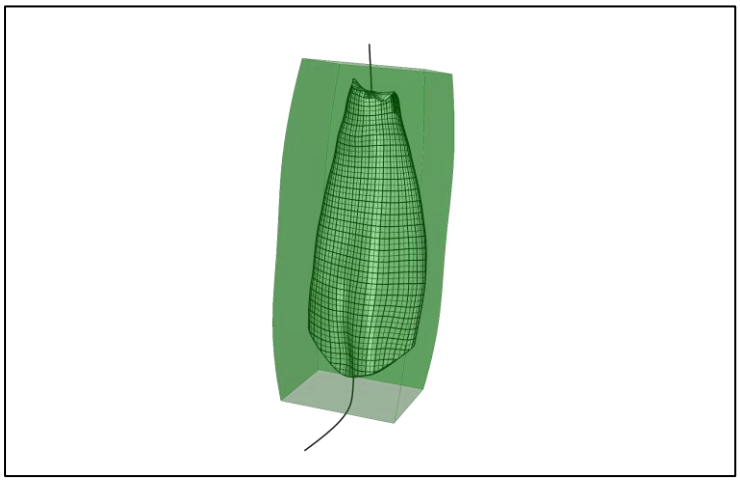
First, in **A**, a cylindrical shape with four surfaces is defined to cover the muscle and define the lattice

coordinates.

The inputs are as follows:

- **Curve**: Reference curve. Use the one from the lattice curve layer.
- **UV**: Coordinate orientation.
- **argDiv**: Number of divisions.
- **rot**: Rotation direction of the coordinates.
- **size_cf, x_size, y_size, decrease**: Parameters for the size of the cylinder.
- **offset, offsetDir**: Offset parameters for the cylindrical shape.

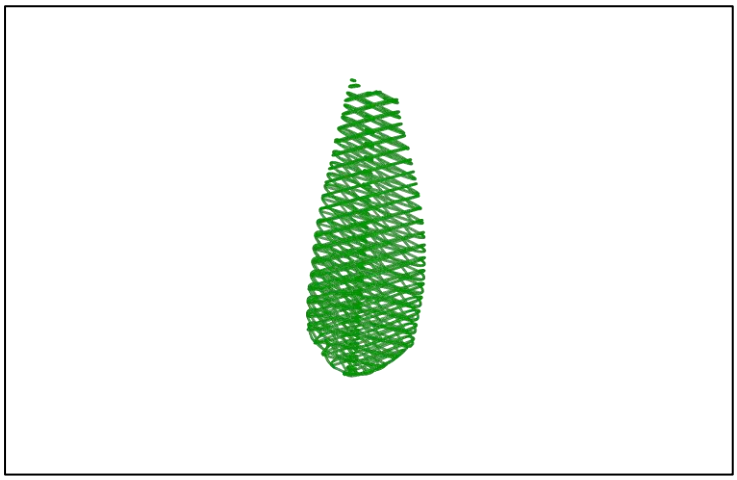
The output **nsrfs** provides cylindrical shapes as the result.



Next, in **B**, the external shape of the muscle and its parameters are specified.
The inputs are as follows:

- **L**: Number of patterns in the contraction direction.
- **X**: Number of patterns in the load-bearing direction.
- **Y**: Number of patterns in the transverse direction.
- **Brep**: External shape of the muscle. Use the one from the muscle_shape_with_wire layer.

In **C**, the following pattern is generated:

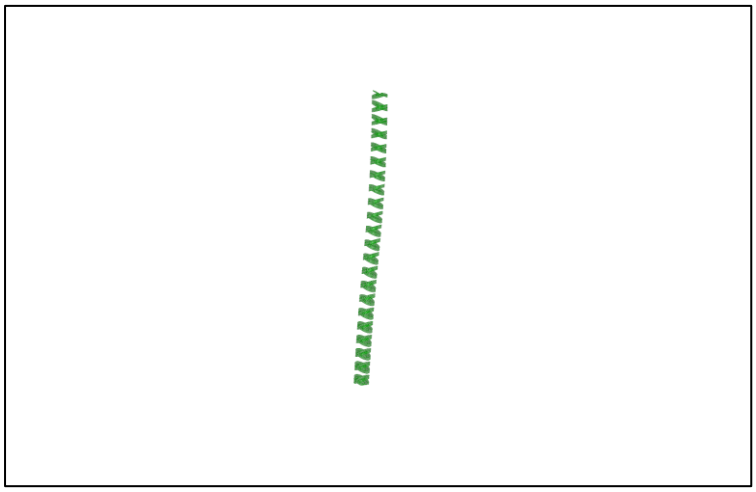


This pattern includes the patterns for both the internal and external structures of the muscle.

Finally, in **D**, the wire path is created.
The inputs are as follows:

- **offset**: Thickness of the path.
- **wires**: Wire curves. Use the ones from the wire_curve layer.
- **d0**, **d1**: Thickness of the path on the inside and outside.
- **shape**: External shape of the muscle. Use the one from the muscle_shape layer.

The output **d** provides the following path.



These patterns can be used as models by using the "Bake" function in Grasshopper.

2. Ligament Generation

To generate ligaments, example geometry inputs are provided in the example.3dm file.

For the generation of ligaments, an example volume input is provided in example.3dm.

Parent layer: under ligament

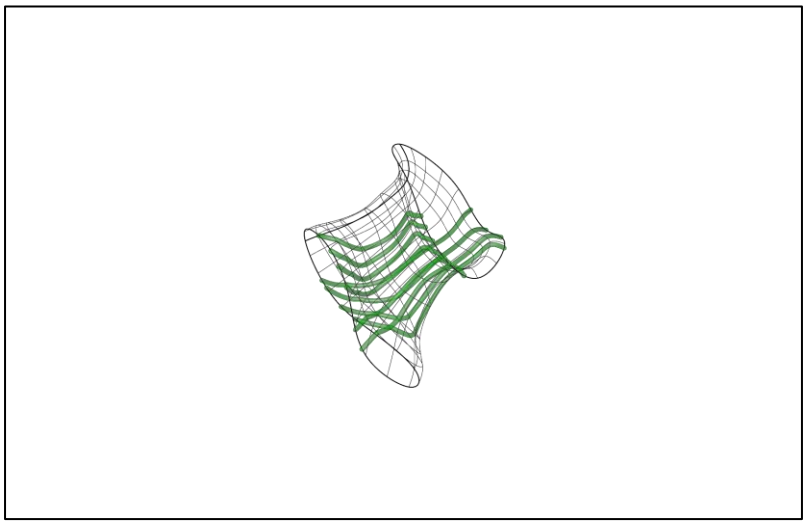
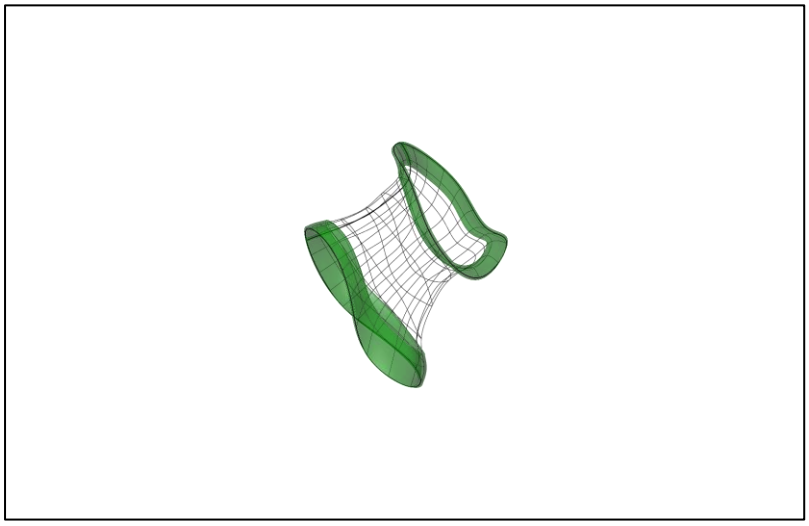
- **Ligament_surface**: Curved surface representing the region where the ligaments exist.
- **Plane_surface**: Corresponding planar surface.
- **Ligament_pattern_curve**: Pattern on the plane.
- **Ligament_pattern_surface**: Pattern on the plane (surface shape).

Section **E** contains the portion for ligament pattern generation. At the top of **E**, there is pattern generation for curves, and at the bottom, there is pattern generation for surfaces.

The pattern generation for curves has the following inputs:

- **Surface_from**: Plane surface. Use the one from the Plane_surface layer.
- **Surface_to**: Ligament surface. Use the one from the Ligament_surface layer.
- **Curve_morph**: Ligament pattern on the plane. Use the one from Ligament_pattern_curve.
- **Transpose, reverse_x, reverse_y**: Parameters related to coordinate orientation.

The following types of ligament patterns are generated.



The pattern generation for surfaces, instead of using the input **curve_morph**, utilizes the surface input **surface_morph** and the **thickness** parameter. The following types of ligament patterns are generated.